

*Delivers centimeter level accuracy within seconds based on U-blox triple-band L1/L2 **ZED-F9P** GNSS RTK module.  
Onboard **RM3100** magnetometer.*

*Datasheet - In Production*



## Features

- 5V - 150mA power supply
- LEDs status: PPS/Power/RTK
- Timepulse & External Interrupt
- USB-C/I<sup>2</sup>C/UART digital interfaces
- Onboard PNI RM3100 compass
- Onboard L1/L2 active antenna

## Applications

- Drones
- RTK Base
- Ground vehicles
- Precise navigation

## Description

The Siderion **Sirius F9P** is an affordable compact and high precision L1/L2 GNSS RTK (Real Time Kinematic) device with an onboard GNSS active antenna.

Based on the last generation of GNSS U-blox ZED-F9P modules, the Sirius F9P offers reliable and fast convergence time to provide centimeter accuracy within seconds.

The device allows concurrent reception of GPS / GLONASS / BeiDou & Galileo signals to improve signal availability.

The Sirius F9P embeds an active L1/L2 GNSS antenna and a professional grade RM3100 magnetometer.

The JST-GH connectors make them perfect to be connected to a Pixhawk3Pro or any other autopilot.

The Sirius F9P GNSS RTK is guaranteed to operate over a temperature range of -40°C to +70°C.

Table 1. Device summary

| Order ref code          | Temperature [°C] | Product size [mm]  | Weight [g] |
|-------------------------|------------------|--------------------|------------|
| SID-GNSS-RTK-SIRIUS_F9P | -40 to +70       | 74.0 x 74.0 x 22.0 | 100        |

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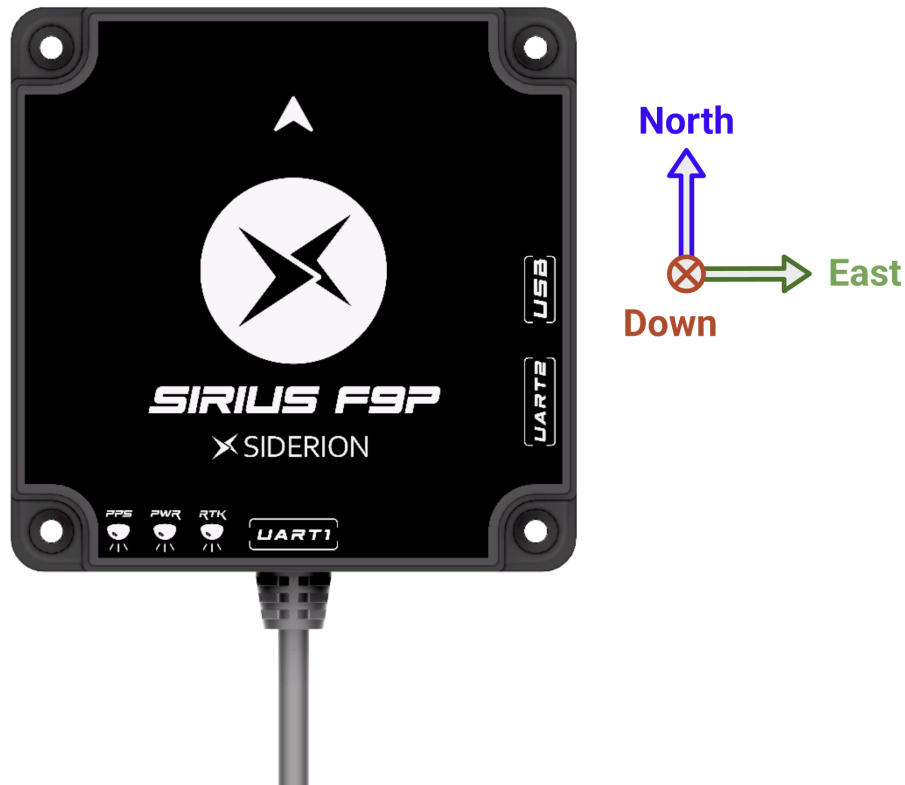
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# 1. Block diagram and pin description

## 1.1 Block diagram

Figure 1. Sirius F9P block diagram & connectivity (top view)



## 1.2 LED description

Table 2. Sirius F9P LED sequence status

| LED name   | Color | Light sequence                                                                      | Comment                              |
|------------|-------|-------------------------------------------------------------------------------------|--------------------------------------|
| Timepulse  | Green |  | Blinking LED at the defined PPS rate |
| Power      | Blue  |  | Solid blue LED when powered ON       |
| RTK Status | Green |  | 3D fix mode / No RTK fix             |
|            |       |  | RTK fix but no FIXED RTK fix         |
|            |       |  | FIXED RTK fix                        |

## 1.3 Pin description

Figure 2. Sirius F9P pinout (top view)

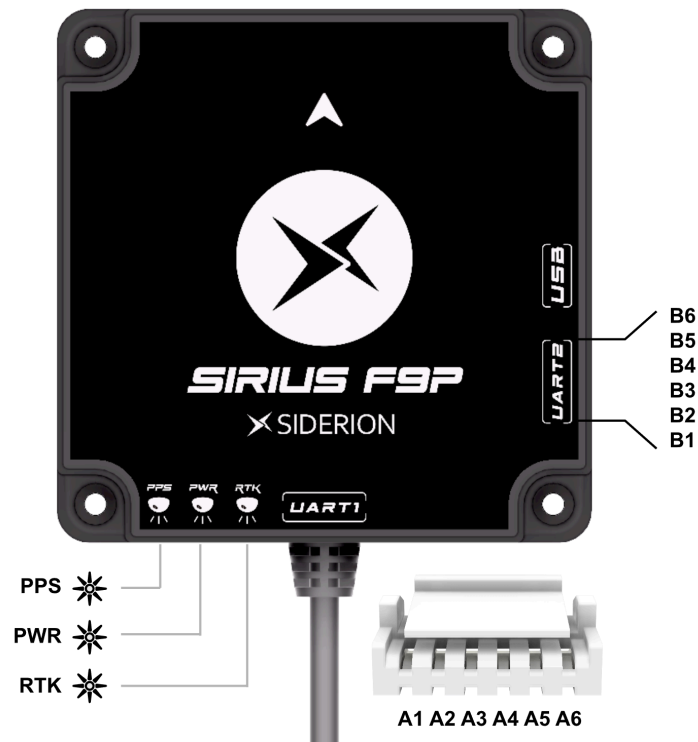


Table 3. Pinout configuration

|      | Pin | Name      | Type | Function                      |
|------|-----|-----------|------|-------------------------------|
| GPS  | A1  | GND       | P    | Ground reference              |
|      | A2  | I2C SDA   | I/O  | I2C data                      |
|      | A3  | I2C SCL   | I    | I2C clock                     |
|      | A4  | UART1 TX  | O    | UART1 transmit                |
|      | A5  | UART1 RX  | I    | UART1 receive                 |
|      | A6  | 5V IN     | P    | 5V input                      |
| UART | B1  | 5V IN     | P    | 5V input                      |
|      | B2  | UART2 TX  | O    | UART2 transmit NMEA/RTCM data |
|      | B3  | UART2 RX  | I    | UART2 receive NMEA/RTCM data  |
|      | B4  | EXTINT    | O    | External interrupt            |
|      | B5  | TIMEPULSE | O    | PPS output                    |
|      | B6  | GND       | P    | Ground reference              |

P : Power / I : Input / O : Output

## 2. Absolute maximum ratings

Stresses above those listed as “absolute maximum ratings” may cause permanent damage to the device. This is a stress rating only and functional operation of the device under these conditions is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

Table 4. Sirius F9P absolute maximum ratings

| Symbol | Parameter               | Maximum value   | Units |
|--------|-------------------------|-----------------|-------|
| Vusb   | USB supply voltage      | -0.3 to +6      | V     |
| Vdd    | Internal supply voltage | -0.5 to +3.6    | V     |
| Vdd_IO | I/O pins supply voltage | -0.5 to Vdd+0.1 | V     |
| Icc_RF | RF output current       | 300             | mA    |
| Prfin  | Input power at RF_IN    | 10              | dBm   |
| W      | Product weight          | 100             | g     |
| TOP    | Operating temperature   | -40 to +70      | °C    |
| TSTG   | Storage temperature     | -40 to +80      | °C    |



This device is sensitive to electrostatic discharge (ESD), improper handling can cause permanent damage to the part.

## 3. Specifications

### 3.1 Ublox ZED-F9P specifications

Table 5. ZED-F9P mechanical and electrical specifications

| Symbol | Parameter                             | Test conditions    | Min.    | Typ.    | Max.     | Units |
|--------|---------------------------------------|--------------------|---------|---------|----------|-------|
| Vusb   | USB supply voltage                    |                    | 4.5     | 5.0     | 5.5      | V     |
| Vdd    | Internal supply voltage               |                    |         | 3.3     |          | V     |
| Vdd_IO | Supply voltage for I/O                |                    |         | 3.3     |          | V     |
| Idd    | Current consumption                   | w/o active antenna |         | 50      |          | mA    |
|        |                                       | w/ active antenna  |         | 130     |          | mA    |
| Vdd_RF | Output RF voltage                     |                    |         | Vdd-0.1 |          | V     |
| Idd_RF | Output RF current                     |                    |         | 50      |          | mA    |
| Vil    | IO pin low lvl input volt             |                    | 0       |         | 0.8      | V     |
| Vih    | IO pin high level input voltage       |                    | 2       |         | Vdd +0.3 | V     |
| Vol    | IO pin low level output voltage       | Iol = 1mA          |         |         | 0.4      | V     |
| Voh    | IO pin high level output voltage      | Ioh = 1mA          | Vdd-0.4 |         |          | V     |
| Top    | Operating temperature                 |                    | -40     |         | +70      | °C    |
| I2Cadd | Default 7bit I <sup>2</sup> C address | 0b0100010 (0x42)   |         |         |          |       |

Table 6. ZED-F9P general performance

| Parameter              | Specifications                   | Value                       |
|------------------------|----------------------------------|-----------------------------|
| Receiver type          | Multi-band GNSS high precision   |                             |
| Accuracy of Timepulse  | RMS                              | 30 ns                       |
| Frequency of Timepulse |                                  | 0.25 Hz to 10 MHz           |
| Operational limits     | Dynamics<br>Altitude<br>Velocity | < 4g<br>50,000 m<br>500 m/s |
| Velocity accuracy      |                                  | 0.03 m/s                    |

Table 7. ZED-F9P performance in different GNSS mode

| GNSS                     | Parameter       | GPS+GLO+GAL<br>+BDS | GPS+GLO           | GPS+BDS           | GPS               |
|--------------------------|-----------------|---------------------|-------------------|-------------------|-------------------|
| Acquisition              | Cold start      | 24 s                | 26 s              | 28 s              | 29 s              |
|                          | Hot start       | 2 s                 | 2 s               | 2 s               | 2 s               |
|                          | Aided start     | 2 s                 | 2 s               | 2 s               | 2 s               |
| Update rate              | RTK             | 8 Hz                | 15 Hz             | 15 Hz             | 20 Hz             |
|                          | PVT             | 10 Hz               | 25 Hz             | 25 Hz             | 25 Hz             |
|                          | RAW             | 20 Hz               | 25 Hz             | 25 Hz             | 25 Hz             |
| Convergence time         | RTK             | < 10 s              | < 10 s            | < 10 s            | < 30 s            |
| Horizontal pos. accuracy | PVT             | 1.5 m CEP           | 1.5 m CEP         | 1.5 m CEP         | 1.5 m CEP         |
|                          | RTK             | 0.01 m + 1ppm CEP   | 0.01 m + 1ppm CEP | 0.01 m + 1ppm CEP | 0.01 m + 1ppm CEP |
| Vertical pos. accuracy   | RTK             | 0.01 m + 1ppm CEP   | 0.01 m + 1ppm CEP | 0.01 m + 1ppm CEP | 0.01 m + 1ppm CEP |
| Sensitivity              | Tracking & Nav. | -167 dBm            |                   |                   |                   |
|                          | Reacquisition   | -160 dBm            |                   |                   |                   |
|                          | Cold start      | -148 dBm            |                   |                   |                   |
|                          | Hot start       | -157 dBm            |                   |                   |                   |

Table 8. ZED-F9P moving-base performance in different GNSS mode

| GNSS             | Parameter | GPS+GLO+GAL<br>+BDS | GPS+GLO | GPS+BDS | GPS     |
|------------------|-----------|---------------------|---------|---------|---------|
| Update rate      |           | 5 Hz                | 8 Hz    | 8 Hz    | 10 Hz   |
| Heading accuracy |           | 0.4 deg             | 0.4 deg | 0.4 deg | 0.4 deg |

## 3.2 PNI RM3100 specifications

Table 9. RM3100 electrical specifications

| Symbol     | Parameter                                                    | Test conditions | Min.     | Typ. | Max.     | Units |
|------------|--------------------------------------------------------------|-----------------|----------|------|----------|-------|
| Vdd        | Internal supply voltage                                      |                 | 2.0      | 3.0  | 3.6      | V     |
| Vdd_IO     | Supply voltage for I/O                                       |                 | 2.0      | 3.0  | 3.6      | V     |
| Vdd_Ripple | Vdd ripple voltage                                           |                 |          |      | 0.05     | Vpp   |
| Idd        | Current consumption                                          |                 |          |      | 1        | mA    |
| Vil        | IO pin low level input voltage                               |                 | 0        |      | 0.3* Vdd | V     |
| Vih        | IO pin high level input voltage                              |                 | 0.7* Vdd |      | Vdd      | V     |
| Iidle      | Idle Mode Current                                            |                 |          |      | 1        | μA    |
| Ileak      | Leakage Current @ DVDD pin<br>(AVDD=AVSS=DVSS=0V, DVDD=3.6V) |                 |          |      | 100      | nA    |
| Top        | Operating temperature                                        |                 | -20      |      | +85      | °C    |
| I2Cadd     | 7bit I2C address                                             | 0b0100000       |          |      |          |       |

Table 10. RM3100 Geomagnetic Sensor Performance

| Parameter                                                                     | Cycle counts |     |     | Units  |
|-------------------------------------------------------------------------------|--------------|-----|-----|--------|
|                                                                               | 50           | 100 | 200 |        |
| Field Measurement Range                                                       | -800 to +800 |     |     | μT     |
| Gain                                                                          | 20           | 38  | 75  | LSB/μT |
| Sensitivity                                                                   | 50           | 26  | 13  | nT     |
| Noise                                                                         | 30           | 20  | 15  | nT     |
| Noise Density @ Max. Single-Axis Sample Rate                                  | 1.2          |     |     | nT/√Hz |
| Repeatability over ±200 μT                                                    | 15           | 8   | 8   | nT     |
| Hysteresis over ±200 μT                                                       | 15           |     |     | nT     |
| Linearity over ±200 μT                                                        | 0.5          |     |     | %      |
| Maximum Single-Axis Sample Rate<br>(divide by 3 for max. 3-axis sample rate)  | 1600         | 850 | 440 | Hz     |
| Single-Axis Average Current @ 24 Hz Sample Rate (equivalent to 3-axis @ 8 Hz) | 70           | 135 | 260 | μA     |
| Bias Resistor, Rb                                                             | 121          |     |     | Ω      |
| Ext. Timing Resistor for Clock, REXT                                          | 33           |     |     | KΩ     |
| Circuit Oscillation Frequency                                                 | 180          |     |     | kHz    |

### 3.3 Embedded L1/L2 GNSS antenna specifications

Table 11. Internal multi-band antenna mechanical and electrical specifications

| Symbol                     | Parameter             | Value                                                           |
|----------------------------|-----------------------|-----------------------------------------------------------------|
| ANTENNA                    |                       |                                                                 |
| Arch                       | Antenna architecture  | Quadrifilar helix                                               |
| Gband                      | GNSS signal bands     | GPS: L1/L2<br>GLONASS: G1/G2<br>BDS: B1/B2<br>Galileo: E1/E2/E5 |
| Gpp                        | Peak gain             | ≥2.0 dBi                                                        |
| Pol                        | Polarization          | RHCP                                                            |
| VSWR                       | V.S.W.R.              | ≤2.0                                                            |
| Acov                       | Azimuth coverage      | 360°                                                            |
| Z                          | Impedance             | 50Ω                                                             |
| LNA                        |                       |                                                                 |
| Glna                       | LNA gain              | 28±2.0dB (Typ. @25°C)                                           |
| Idd                        | Operating current     | ≤50mA                                                           |
| Nfig                       | Nois Figure           | <1.5dB                                                          |
| Pbf                        | Pass band fluctuation | ±1.5dB                                                          |
| MECHANICAL & ENVIRONMENTAL |                       |                                                                 |
| IP                         | IP rating             | IP66                                                            |
| Top                        | Operating temperature | -40°C to +85°C                                                  |
| Size                       | Antenna dimensions    | 60.0 x 60.0 x 20mm                                              |

## 4. Applications

## 5. Communication interfaces

There are several communication interfaces, including UART, I2C, and USB. All inputs have internal pull-up resistors in normal operation and can be left open if not in use. All the PIOs are supplied by VCC, therefore all the voltage levels of the PIO pins are related to Vdd supply voltage.

### 5.1 UART Interface

There are two UART interfaces: UART1 and UART2. UART1 and UART2 operate up to and including a speed of 1 Mbaud. No hardware flow control on UART1 and UART2 is supported.

Figure 3. UART chronogram specifications

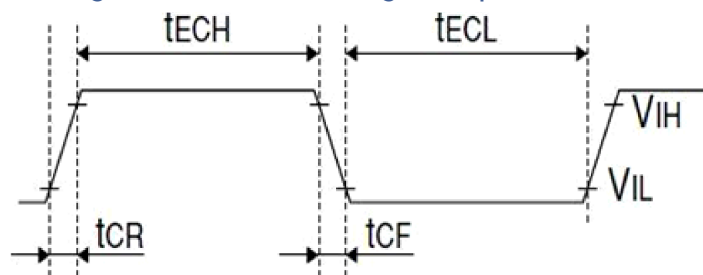


Table 12. ZED-F9P serial UART timing specifications

| Symbol | Parameter                          | Min.    | Max.    | Unit |
|--------|------------------------------------|---------|---------|------|
| Vil    | LOW-LEVEL input voltage            | 0       | 0.2xVdd | V    |
| Vih    | HIGH-LEVEL input voltage           | 0.7xVdd | Vdd+0.3 | V    |
| tECH   | HIGH period of external data input | 0       | 0.4     | μs   |
| tECL   | LOW period of external data input  | TBA     | TBA     | μs   |
| Ru     | Baudrate                           | 9600    | 921600  | bps  |
| tCR    | Rise time of data                  |         | 5       | ns   |
| tCF    | Fall time of data                  |         | 5       | ns   |

## 5.2 Slave I2C interface

An I2C compliant interface is available for communication with an external host CPU. The interface can be operated in slave mode only. It is fully compatible with Fast-Mode of the I2C industry standard. Since the maximum SCL clock frequency is 400 kHz, the maximum bit rate is 400 kbit/s. The interface stretches the clock when slowed down while serving interrupts, therefore the real bit rates may be slightly lower.

Figure 4. I2C chronogram specifications

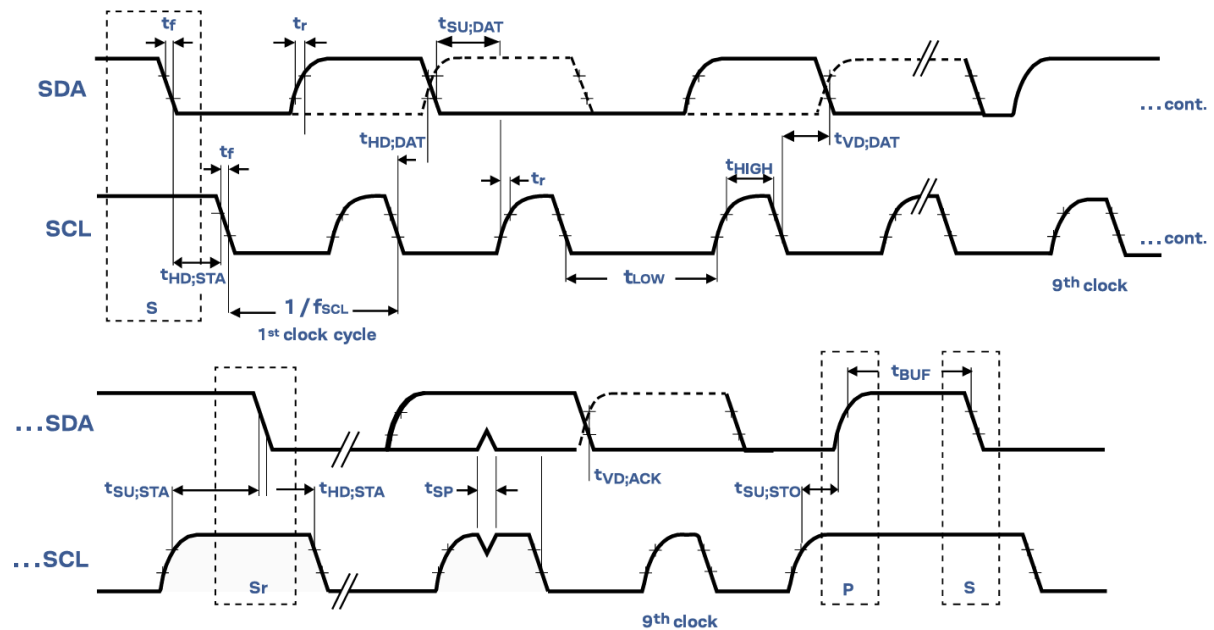


Table 13. Sirius F9P I2C timing specifications

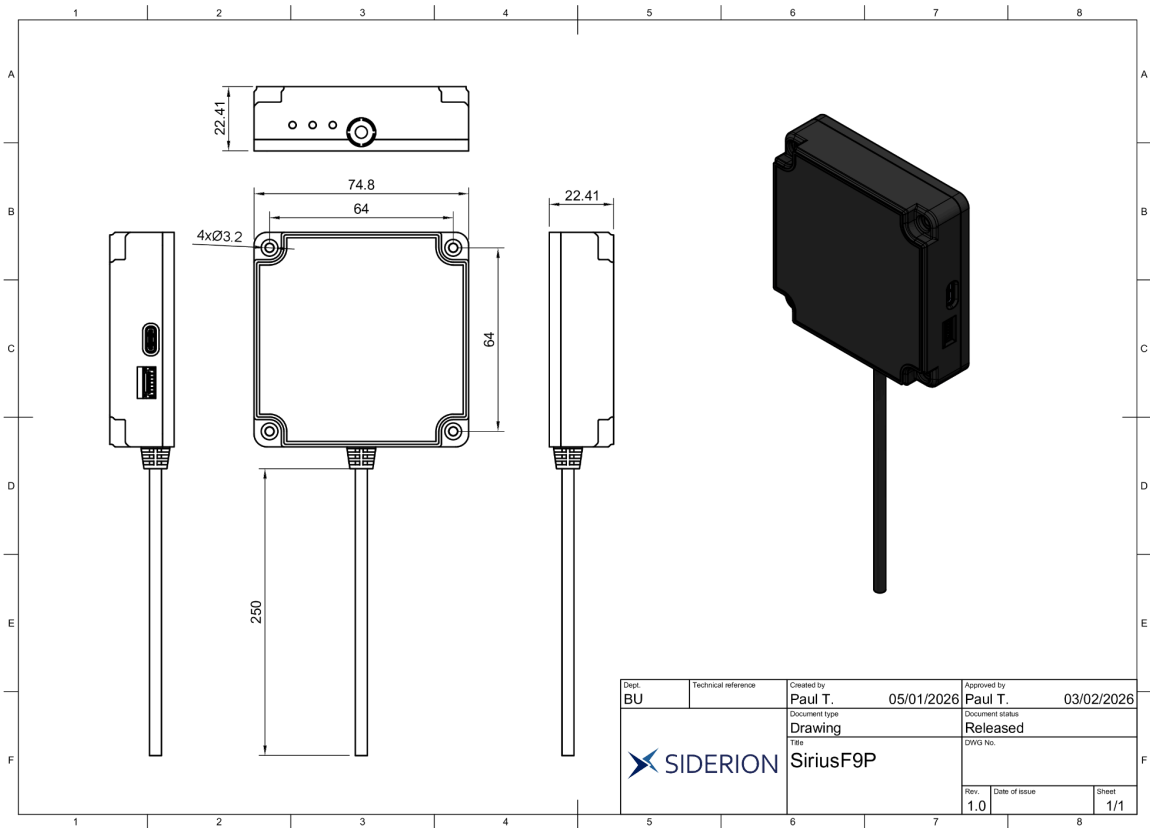
| Symbol | Parameter                 | Min     | Max     | Unit |
|--------|---------------------------|---------|---------|------|
| Vil    | LOW-LEVEL input voltage   | Vss-0.3 | 0.3xVdd | V    |
| Vih    | HIGH-LEVEL input voltage  | 0.7xVdd | Vdd+0.3 | V    |
| Vol    | LOW-LEVEL output voltage  |         | 0.4     | V    |
| Voh    | HIGH-LEVEL output voltage | Vdd-0.4 |         | V    |
| Fscl   | SCL clock frequency       | 0       | 400     | KHz  |

## 5.3 USB interface

A USB type C interface, which is compatible with USB version 2.0 FS (Full Speed, 12 Mbit/s), can be used for communication as an alternative to the UART.

6. Mechanical drawings

Figure 5. Sirius F9P v2.1 mechanical drawings



## 7. Revision history

Table 14. Document revision history

| Date        | Revision | Changes                                 |
|-------------|----------|-----------------------------------------|
| 16-Feb-2026 | 1.0      | SiderionDoc_SiriusF9P / Initial release |

## 8. Appendix

U-blox ZED-F9P datasheet : [ZED-F9P-05B Data sheet](#)

U-blox ZED-F9P integration manual : [ZED-F9P Integration manual](#)

U-blox ZED-F9P interface description : [u-blox F9 HPG 1.51 Interface description](#)

Siderion user's guide:

Siderion tutorials:

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